Residue Arithmetic's using Reversible Logic Gates

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***Abstract*—The Residue number system (RNS) has been employed for efficient parallel carry-free arithmetic computations in DSP applications. Residue addition is the instrumental component in implementing residue converters and channels in RNS. On the other side Reversible Logic is becoming one of the potential power optimization techniques in Low Power CMOS design. In this research paper we have proposed CMOS implementation of two different reversible logic architectures for 4-bit generic modulo-m ripple residue adder using 4×4 TSG , DPG and 3×3 Fredkin Reversible logic gates and one architecture for 4-bit generic modulo-m carry look ahead residue adder using 4×4 RMF and 3×3 Fredkin Reversible logic gates along with two irreversible logic based 4-bit generic modulo-m residue adder one for irreversible 4-bit generic modulo-m ripple residue adder and another for 4-bit generic modulo-m carry look ahead adder. Proposed architectures are analyzed in terms of power, delay, garbage o/p, constant inputs and transistor count using 180nm technology node at 1.8 v with operating frequency of 200 MHz. It is observed that DPG based Reversible Residue Ripple adder has 40% more efficient than irreversible Residue Ripple adder and RMF based Reversible Residue CLA adder has 33% more efficient than irreversible Residue CLA added. The implementation is based on Reversible pass transistor Logic (R-CPL).**

Keywords-Residue Number system, Reversible logic gates Residue adder, CMOS Technology, Reversible Pass transistor Logic (R-CPL).

I. INTRODUCTION

The computation carried out in conventional computers is irreversible, i.e., once a logic block generates the output bits, the input bits are lost. This loss of input bits leads to the power consumption in the system.Landauer [1] has shown that for every bit of information lost in logic computations that are not reversible, kT×log2 joules of heat energy is generated, where k is the Boltzmann's constant, and T is the absolute temperature at which the computation is performed. The energy dissipation due to bit loss is not significant for the current clock speed of computers. However, as the clock speeds of computers are elevated, the frequency with which the information bits are lost will also increase. Reversible logic circuits avoid energy loss by "uncomputing" the computed information, thereby recycling the energy in the system [26]. Bennett [2] showed that zero power dissipation in logic circuits is possible only if a circuit is composed of reversible logic gates. A gate is reversible if there is a distinct output assignment for each distinct input [2]. Thus, a reversible gate's inputs can be uniquely determined from its outputs. A reversible logic gate must have the same Number of inputs and outputs [2]. Reversible gates are balanced, i.e., the outputs are 1s for exactly half of the inputs. A circuit composed of reversible gates and without constants on its inputs realizes only balanced functions. It can realize non balanced functions only with garbage output.

A reversible logic gate is an n-input, n-output logic device with one-to-one mapping. Reversible circuits are constructed using reversible logic gates. These reversible circuits not only produce unique output vector from each input vector but also the input can be reconstructed from the outputs. A reversible circuit should be designed using a minimum number of reversible gates. Fan-out and loops are not allowed in reversible logic circuits [3].

Interest in reversible logic is sparked by its applications in several technologies, such as quantum [14], CMOS [25], optical and nanotechnology. Reversible implementations are also found in thermodynamics and adiabatic CMOS Reversible logic is an emerging research area.

Residue Number system is a carry-free arithmetic system that enhance the speed of computations, and exhibits fault-tolerant capabilities [7].Addition, subtraction and multiplication are carried out on residue digits independently and concurrently [8].Independence of residue digits eliminates complex VLSI interconnection patterns among different logic components, offers functional modularity, and leads to simultaneous operations on all digits.RNS has shown significant efficiency in implementing different types of digital signal processing applications. It has been successfully implemented in applications involving the design of digital filters [8].Therefore, designing an efficient residue adder is an important task in realizing different RNS-based applications[7],[8].Moreover, residue adders are very important components in building residue based- multiplier, residue to binary converters, binary to residue converters and other arithmetic operations.

The RNS is a non-weighted integer number system that is defined by an N-tuple base of pair wise relatively prime Positive integers$\{m_0, m_1, m_2, \ldots\ m_{N-1}\}$, which are collectively known as the moduli of the system[8].For a given base, an integer X is represented by an N-tuple word,$\{x_0, x_1, x_2, \ldots\ x_{N-1}\}$,where $x_i = |X|_{m_i}$,i.eit is the non negative remainder when dividing$X$ by the modulus m_i .Let $\diamond$ denote the binary operation of addition,subtraction, or multiplication. It then follows that $Z = X \diamond Y$ is Isomorphic to $= \{z_0, z_1, z_2 \ldots\ z_{N-1}\ \}$,where $z_i = |x_i \diamond y_i|_{m_i}$, $i \in [0, N-1]$.Note that z_i is solely dependent upon $x_i$and y_i.And, hence, the arithmetic operation $\diamond$ is performed in parallel with no interaction between the RNS channels.

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As a direct consequence of this property, RNS systems are capable of performing high-speed addition and multiplication, usually at a fraction of the time taken in traditional two's complement systems. In addition to high-speed implementations, RNS circuits have also demonstrated power efficiency when implementing digital signal processing (DSP) functionality [9].The minimization of power dissipation has become an important performance objective as the trend increases towards portability, denser circuits, and high performance.

This paper is organized as follows: Section II describes the background of Reversible logic gates. Section III explains the CMOS dual pass logic implementation of Feynman, TSG, DPG, RMF and Fredkin Gate .Section IV gives brief about Residue adders. Sections V describe the proposed generic modulo-m residue ripple & carry look ahead adders using reversible logic gates. In Section VI comparative Result analysis has been done followed by Section VII which concludes the work done.

II. REVERSIBLE LOGIC GATES

A Reversible gate is a n-input, n-output (denoted by $n \times n$) circuit that produces a unique output pattern for each possible input pattern. Reversible gates are circuits in which the number of outputs is equal to the number of inputs and there is a one to one correspondence between the vector of inputs and Outputs, i.e., it can generate unique output vector from each input vector and vice versa. The complexity and performance of the circuit is decided on the following parameters.

Garbage outputs: The number of unused outputs present in the reversible logic circuit.

Number of reversible gates: Total number of reversible gates used in the circuit.

Delay: Maximum number of unit delay gates in the path of propagation of inputs to outputs. It represents the total number of reversible gates used between the primary inputs and the outputs of a reversible logic circuit.

Constant inputs: The number of inputs which are maintained constant at 0 or 1 in order to get the required function. They are necessary to synthesize a reversible function.

Quantum cost: The number of 1×1 or 2×2 reversible logic gates used in the quantum equivalent of the reversible circuit.Every reversible gate can be calculated in terms of quantum cost and hence the reversible circuits can be measured in terms of quantum cost. Reducing the quantum cost from reversible circuits always a challenging one and works are still going on in this area

An $n \times n$ reversible gate can be represented as $I_v = (I_1, I_2, I_3, \ldots I_n)$, $O_v = (O_1, O_2, O_3, \ldots O_n)$, Where I_v and O_v are input and output vectors respectively. Several reversible logic gates have been proposed in the past years. There exist many universal reversible gates [10-13].

Some of the popular and important gates are introduced:

***NOT gate*:** There is only one 1×1 gate in reversible logic, which isinverter shown on figure 1. The input vector is $I_v = (A)$ and the output vector is $O_v = (P = \bar{A})$.This gate is very important since it does not introduce garbage outputs and the quantum cost is 0.



Figure 1:1×1 NOT Gate

***Feynman Gate*:** Figure 2 shows a 2×2 Feynman gate [14].The input vector is $I_v = (A, B)$ and the output vector is $O_v = (P = A, Q = A \oplus B)$. Quantum cost of a Feynman gate is 1.



Figure 2:2×2 Feynman Gate

***Toffoli Gate*:** Figure 3 shows a 3×3Toffoli gate [15].The input vector is $I_v = (A, B, C)$ and the output vector is $O_v = (P = A, Q = A \oplus B, R = AB \oplus C)$.Quantum cost of a Toffoli gate is 5.

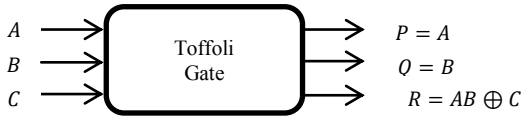
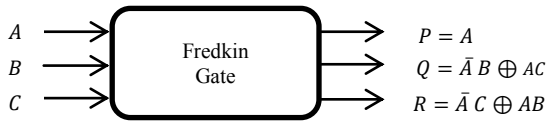
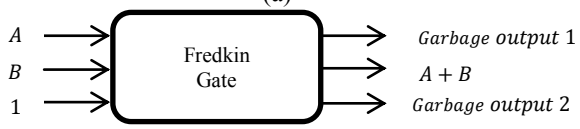


Figure 3: 3×3 Toffoli gate

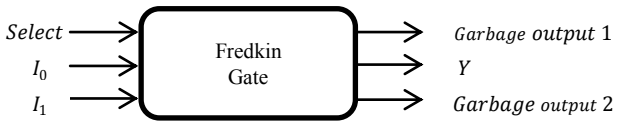
***Fredkin Gate*:** Figure 4(a) shows a 3×3 Fredkin gate [16]. The input vector is $I_v = (A, B, C)$ and the output vector is $O_v = (P = A, Q = \bar{A}\,B \oplus AC, R = \bar{A}\,C \oplus AB)$ .Figure 4(b) shows the implementation of OR gate using Fredkin gate and Figure 4(c) shows the implementation of 2:1 Multiplexer using Fredkin gate. Quantum cost of a Fredkin gate is 5.



(a)



(b)



(c)

Figure 4: (a)3×3 Fredkin gate(b)As OR gate(c) As 2:1Mux

***Peres Gate*:** Figure 5 shows a 3×3 Peres gate [10]. The input vector is $I_v = (A, B, C)$ and the output vector is $O_v = (P = A, Q = A \oplus B, R = AB \oplus C$. Quantum cost of a Peres gate is 4.

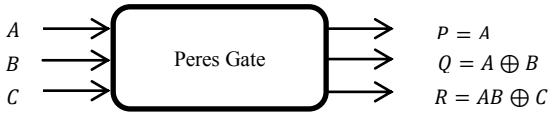


Figure 5:3×3 Peres gate

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Double Peres gate: Figure 6(a) shows a 4× 4 Double Peres Gate. The input vector is $I_v = (A, B, C, D)$ and the output vector is $O_v = (P = A, Q = A \oplus B, R = A \oplus B \oplus D, S = (A \oplus B)D \oplus AB \oplus C$ .The output is defined by P =A,Q=A⊕B,R=A⊕B ⊕D and S=(A⊕B)D⊕AB⊕C. The full adder using DPG is obtained with $C = 0$ and $D = C_{in}$ and sum and carryout is available at R and S outputs respectively shown in figure 6(b). Its quantum cost is equal to 6.

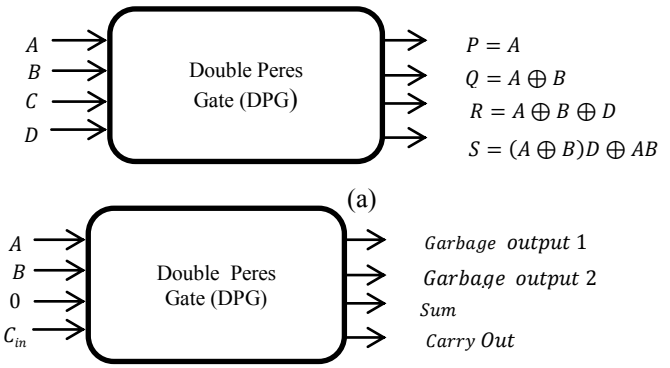


Figure 6: (a) DPG Gate(b)DPG as Full Adder

***TSG Gate*:** Figure 7(a) shows a TSG gate. The input vector is$I_v = (A, B, C, D)$ and the output vector is $O_v = (P = A, Q = \bar{A}\bar{C} \oplus \bar{B}, R = (\bar{A}\bar{C} \oplus \bar{B}) \oplus D, S = (\bar{A}\bar{C} \oplus \bar{B}) \oplus D \oplus AB \oplus C$.The full adder using TSG is obtained by$C = 0\ and\ D = C_{in}$which is shown in figure 7(b).Its quantum cost is equal to 6[17]

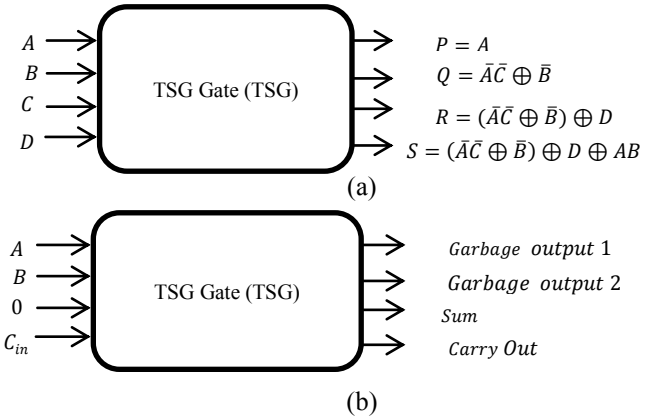
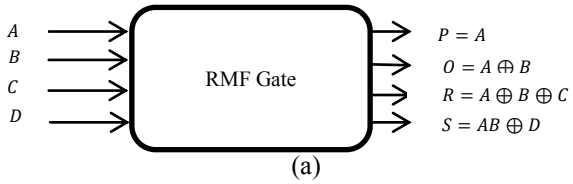


Figure 7:(a)TSG Gate (b)TSG as Full adder

***RMF Gate*:** Figure8(a) shows a RMF gate. The input vector is$I_v = (A, B, C, D)$ and the output vector is $O_v = (P = A, Q = A \oplus B, R = A \oplus B \oplus C, S = AB \oplus D$.The key consideration while designing reversible carry look ahead adder is to generate P_i,S_i and G_i with minimum garbage outputs .RMF is abile to realize the P_i,S_i and $\bar{G}_i$ with low bound of 1 reversible gate and 1 garbage output which can be obtained by $D =$ 1which is shown in figure 8(b)[27]



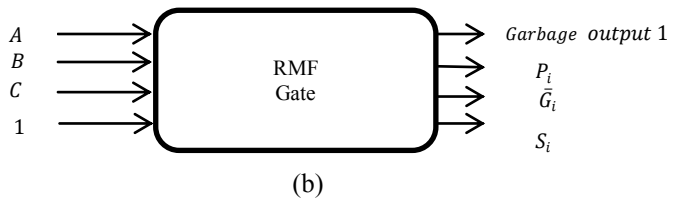


Figure 8: (a) RMF Gate (b)RMF as Modified Full Adder

***TKS Gate*:** Figure 9 shows a TKS gate. The input vector is $I_v = (A, B, C)$ and the output vector is $O_v = (P = A\bar{C} + BC, Q = A \oplus B \oplus C, R = AC + B\bar{C})$ .TKS and Pares gate combination is used for generating multi-input NAND Gate.

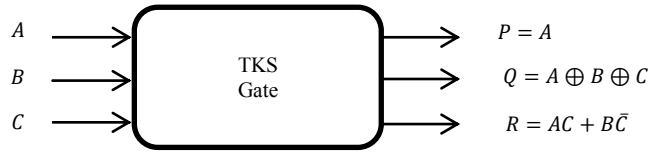


Figure 9:TKS Gate

III. CMOS REALIZATION OF REVERSABLE LOGIC GATES

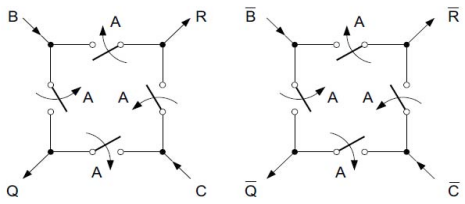
Conventional Static CMOS has been a technique of choice in most Processor design. Standard CMOS is the most common and widely utilized digital logic in almost any application field. Still, other digital logic styles exist and are utilized in the industry as well. Since we aim in this Paper work for special design characteristics, such as very low power consumption, it is fair to have a look at other digital logic design styles as well.

Complimentary pass logic CPL [28] uses only an N-MOSFET network for the implementation of logic functions, thus resulting in low input capacitance and high-speed operation. Because the high voltage level of the pass transistor outputs is lower than the supply voltage level by the threshold voltage of the pass transistors, the signals have to be amplified by using CMOS inverters at the outputs. CPL has been used in conventional computers from many years in order to improve special circuits such as static RAM and other circuits that use XOR'[s]. This was used for Reversible logic gates by[5]. This logic family is called Reversible pass transistor Logic (R-CPL).

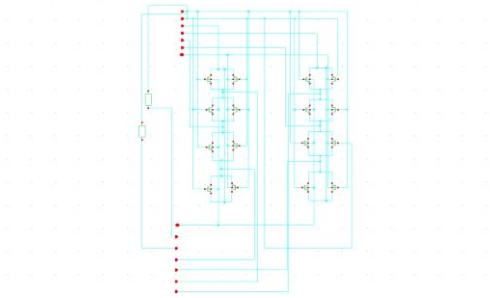
The schematics are designed with Cadence Design frame work II EDA full custom software. The channel width of the transistors is 2µm for the NMOS and 50µm for the P-MOS and channel length is 0.18µm with operating frequency is 200Mhz.For the implementation of On-Off switch we have used CMOS Transmission gate which consists of N-MOS and P-MOS Transistors connected parallel. Gates are designed as controlled rings for opening or closing the switch. We stress that these circuits have no power supply inputs,thouse there are neither V_{dd} or V_{ss} nor ground .all energy provided at the output originate from the inputs

In this proposed work this R-CPL logic is utilized for implementing DPG, TSG, RMF and Fredkin gates.. Figure 10,11,12 and 13shows switch representation and Transistor schematics of captures using Virtuoso Cadence Tools for , Fredkin gate ,DPG, TSG, RMF.

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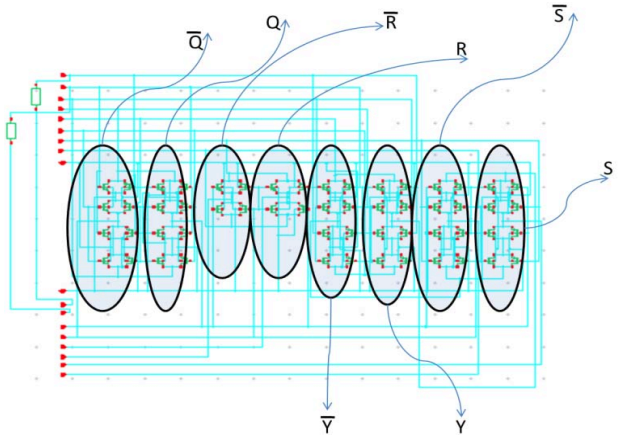
(a)



(b)

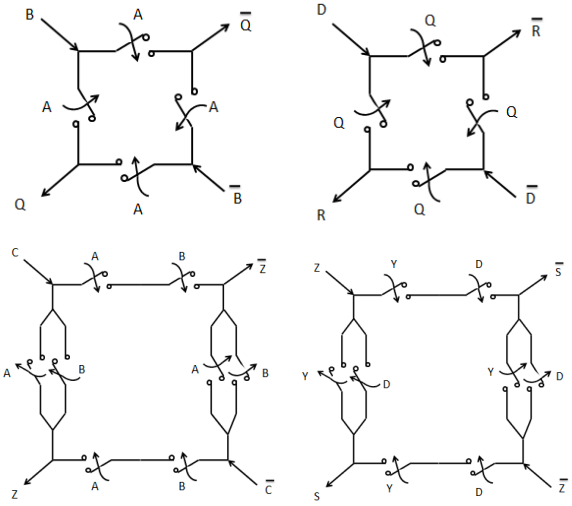
Figure 10:(a) Switch representation
(b) Transistor implementation of Fredkin gate

(a)



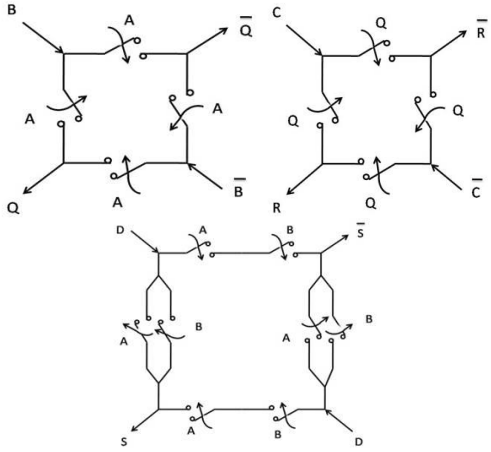
(a)

Figure 11: (a) Transistor implementation of TSG gate



(a)

Figure 12: (a) Switch Representation of DPG gate



(a)

Figure 13: (a) Switch Representation of RMF gate

IV. RESIDUE ADDERS

Residue adders are fundamental arithmetic components typically employed in residue number system (RNS)-based digital signal processing (DSP) systems. In general, the design approach for modular adders falls into two distinct categories. On the one hand, they can be designed for flexibility in which case the methodology allows the design of adders for any moduli which we call as generic modulo-m adder proposed in [20], [21] . On the other hand, if one desires architectural simplicity and the best performance, then adders designed for a specific set of efficient moduli have been proposed previously [18], [19].In this paper we are proposing reversible generic moduli-m residue ripple adder and CLA adder . The residue sum of two residue digits, $\langle X+Y\rangle \, mod\, M$, is the residue of the sum $X+Y$ with respect to the modulus m. The operation may be defined as follows.

$$\langle X+Y\rangle_m = \begin{cases} X+Y, & X+Y < m \\ X+Y-m, & X+Y \geq m \end{cases}$$

Bayoumi [20] described a modulo adder that utilizes two) binary adders and a multiplexer as shown in Figure 14. The first adder adds A and B .the second one adder sum to $2^n - m$ which is nothing but 2's complement of m. The carry bit generated by the first and second adder indicates whether or not $A+B$ is greater than m. If anyone carry out of the two adders is 1 the second adder sum is the result otherwise first adder result is the final result.

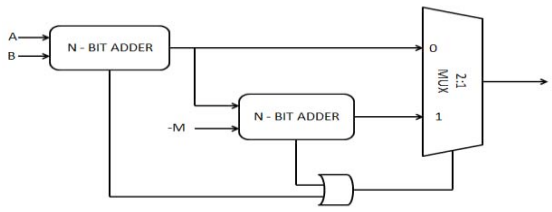


Figure 14: Bayoumi [20] Residue Adder

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V. PROPOSED REVERSABLE GENERIC MODULO-M ADDERS

Our proposed Generic 4-bit modulo–m Residue ripple adder designs have utilized TSG,DPG & Fredkin gates and Generic 4-bit modulo–m Residue CLA adder designs have utilized RMF,Feynman ,TKS, Peres & TSG gates. The basic cell for such a residue adder is a full adder (FA) accepting three bits. As proposed in[22],[23],and [24] a full adder constructed using TSG and DPG has been considered for the design of Reversible generic modulo-m ripple adder. Bayoumi [20] described modulo adder is utilized in this proposed architectures .Two different residue ripple adder architectures are proposed using the combination of DPG and TSG gate. And also one residue CLA adder architecture is proposed.

A. Proposed Architecture 1:

DPG & Fredkin based Reversible 4-bit residue ripple adder

In this Architecture DPG gate is used as a full adder cell and Fredkin gate is used to implement the OR gate and four 2:1 MUX. The Figure 13 shows the architecture of DPG based Reversible residue adders.

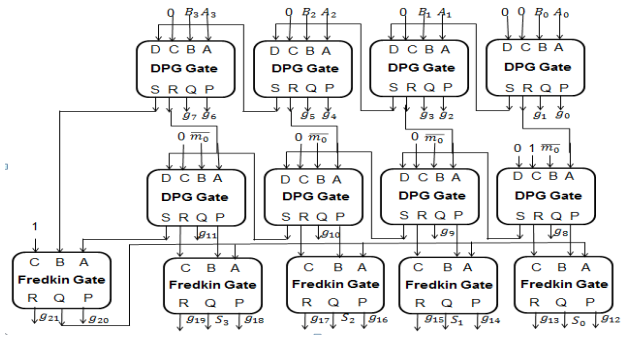


Figure13:DPG& Fredkin based Reversible 4-bit residue adder

B. Proposed Architecture 2:

TSG & Fredkin based Reversible 4-bit residue ripple adder

In this proposed architecture TSG gate is used as a full adder cell as shown in figure 6(b) and Fredkin gate is used to implement the OR gate and 2:1 MUX. The figure 14 shows the architecture of TSG based Reversible residue adders.

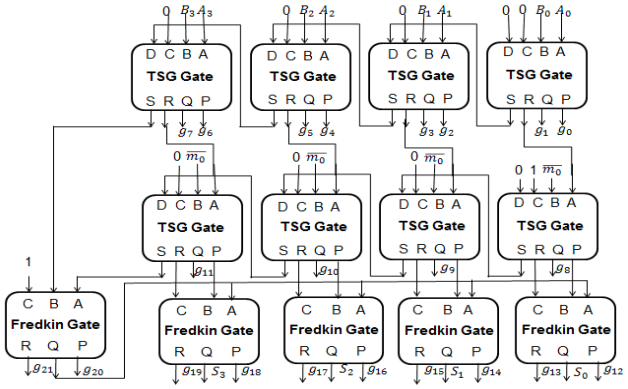


Figure14:TSG & Fredkin based Reversible 4-bit residue ripple adder

C. Proposed Architecture 3:

RMF based Reversible 4-bit residue CLA adder

The key values needed for carry look ahead logic are propagate and generate signals along with sum. Instead of choosing full adder, modified full adder is utilized[29]. RMF gate can realize Propagate p_i , generate $\overline{g_i}$ and sum s_i with lower bound of 1 reversible gate and one garbage bit.4-bit carry look ahead adder using RMF proposed by [27] is utilized for constructing two n-bit binary adder. Figure 15 show the CLA adder architecture proposed by [27].Peres gate is used for generating NAND function. Feynman gate is used for NOT function. RMF is used for modified full adder. TKS is used for multi-input NAND function..Figure 16 shows the Proposed RMF based Reversible 4-bit residue CLA added. Fredkin gate is used for OR function and 2:1 Mux operation

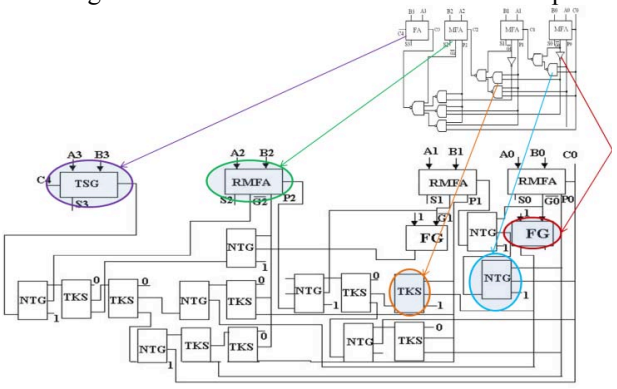


Figure 15:4-bit CLA binary adder[27]

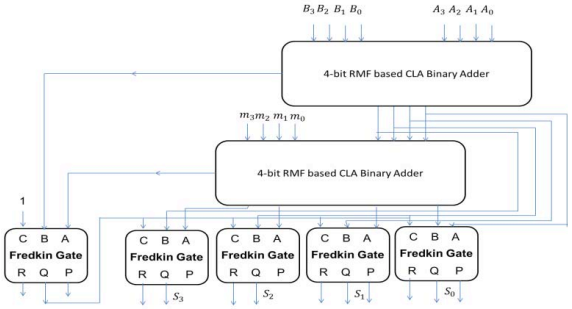


Fig 16:RMF based Reversible 4-bit residue adder.

VI. RESULTS AND ANALYSYS

The 4-bit Revisable Residue ripple &carry look ahead adder is compared with irreversible Residue ripple & carry look ahead adder based on the performance parameters like no of garbage outputs, No of constant inputs, number of gates, propagation delay, number of transistors, power dissipation and power product delay . To achieve better power performance, the circuits were designed using CMOS complementary pass transistor logic using Reversible logic process. The channel width of the transistors is 2µm for the NMOS and 50µm for the PMOS and channel length is 0.18µm with operating frequency is 200Mhz.Cadence Design Frame work II custom design EDA software tool is chosen for schematic implementation. Table 1 shows the 4-bit Reversible Residue Adder Comparative Analysis.

From comparative analysis it has been observed that Reversible generic modulo-m Residue adder designed using DPG along with Fredkin gate (Architecture-I) is 16% efficient than implementation with TSG gate and Fredkin gate =

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(Architecture-II) and 40% efficient than Irreversible generic residue Ripple Carry adder with respective to Power delay Product. RMF based Residue adder (Architecture-3) is superior than irreversible generic Residue CLA adder by 33% .Architecture-1 consumes lowest power among all the three reversible architectures.

Table 1: 4-bit Residue Adder Comparative Analysis

(Operating conditions V_{DD}=1.8v, C_L=100fF,fclk= 200MHz,180 nm)

Residue Adder	*No of Garbage O/P's*	*No of Constant I/P's*	*No of Gates*	*Propagation Delay [ns]*	*No of Transistors*	*Power Dissipation [nW]*	*Power Delay Product *10^{-18}*
Arch-1 (DPG+FG) Revisable Ripple carry	**22**	**8**	**13**	**30.8**	**444**	**0.9014**	**27.76**
Arch-2 (TSG+FG) Revisable Ripple carry	**22**	**8**	**13**	**31.08**	**528**	**1.487**	**46.21**
Arch-4 (RMF) Revisable CLA	**73**	**12**	**22**	**22.36**	**686**	**1.923**	**42.99**
Arch-5 Irreversible Ripple Carry	**NA**	**NA**	**NA**	**31.29**	**244**	**1.642**	**51.37**
Arch-6 Irreversible CLA	**NA**	**NA**	**NA**	**26.36**	**288**	**2.2**	**57.99**

VII. CONCLUTION

The focus of the work described here centered on finding good architectures for reversible generic modulo-m residue adder. It is observed that DPG based Reversible Residue Ripple adder has 40% more efficient than irreversible Residue Ripple adder and RMF based Reversible Residue CLA adder has 33% more efficient than irreversible Residue CLA adder. In future, the design can be extended to more efficient reversible residue adder using CMOS adiabatic logic and special class of residue adders.

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